Kolmogorov Smirnov Tests - Lab Activity - Example Solution

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This lab activity will help you practice your skills with Kolmogorov Smirnov (KS) tests.

Randomly Generated Comparisons

```r
n <- 5 # number of observations
X <- rnorm(n, 25, 2)
Y <- rgamma(n, 5, 1/5) # gamma dist with same mean as normal
ks.test(X,Y)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  X and Y
## D = 0.6, p-value = 0.3571
## alternative hypothesis: two-sided
```

Here, the two distributions have the same mean, but one is bell-shaped and the other right-skewed.

For a sample of size 5 from each distribution, do you detect a difference between them? You can re-run the code several times to see if the answer changes. (Note we did not set a seed here - everyone should get different random samples. If you want to get the same repeated results, set a seed.)

SOLUTION:

The sample of size 5 seems to be too small for us to detect a difference in distributions. We ran the test several times and obtained p-values of 0.3571, 0.07937, 0.873, 0.07937, and 0.873.

Now, try increasing n. For what values of n can you fairly confidently tell that X and Y are coming from different distributions?

SOLUTION:

```r
# Use the R code provided above,
# but now increase n from 5 to larger values of interest to you.

n <- 10 # number of observations
X <- rnorm(n, 25, 2)
Y <- rgamma(n, 5, 1/5) # gamma dist with same mean as normal
ks.test(X,Y)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  X and Y
## D = 0.5, p-value = 0.1678
## alternative hypothesis: two-sided
```

At n=10, we obtained p-values of 0.4175, 0.05245, 0.002057, and 0.4175. Thus, we only detected the difference one out of four times, assuming an alpha of 0.05. Answers will obviously vary from student to student.

With an n=15, out of 5 tests, we determined there were different distributions 3 out of 5 times. With n=25, we fairly confidently (only 1 out of 10 tests did not reject the null) can tell that there are two distributions here. So, we clearly want larger sample sizes. Note: the two groups don't need to have equal sample sizes either.

Suppose we think X comes from a normal distribution. How far off do we need to be when guessing the mean and SD to say that we do not have sufficient evidence that the sample comes from that distribution? Is it worse to be far off guessing the mean or the SD?

```
X <- rnorm(50, 25, 2)
favstats(~ X) # see how sample stats look compared to theoretical dist
```

```
##      min       Q1   median       Q3     max     mean       sd  n missing
##  19.5384 23.54337 24.60095 26.24497 28.3047 24.66454 2.040589 50       0
```

```
ks.test(X, "pnorm", 25, 2) # original guess from knowledge you will not normally have
```

```
##
##  Exact one-sample Kolmogorov-Smirnov test
##
## data:  X
## D = 0.12095, p-value = 0.4239
## alternative hypothesis: two-sided
```

```
ks.test(X, "pnorm", 10, 5) # bad guess
```

```
##
##  Exact one-sample Kolmogorov-Smirnov test
##
## data:  X
## D = 0.97178, p-value = 8.882e-16
## alternative hypothesis: two-sided
```

Try different values for the mean and SD to get a sense of when the test starts rejecting the null hypothesis. Summarize what you find.

SOLUTION:

Note: Our listed results may differ from the rendered ones. We could set a seed to avoid this, but you probably already have different results too.

```
ks.test(X, "pnorm", 23, 5) # guess - detected a difference
```

```
##
##  Exact one-sample Kolmogorov-Smirnov test
##
## data:  X
## D = 0.37001, p-value = 1.193e-06
## alternative hypothesis: two-sided
```

```
ks.test(X, "pnorm", 23, 2) # guess - detected a difference
```

```
##
##  Exact one-sample Kolmogorov-Smirnov test
##
## data:  X
## D = 0.38228, p-value = 4.422e-07
```

```
## alternative hypothesis: two-sided
```

```r
ks.test(X, "pnorm", 24, 2) # guess - detected a difference
```

```
##
##  Exact one-sample Kolmogorov-Smirnov test
##
## data:  X
## D = 0.18487, p-value = 0.05727
## alternative hypothesis: two-sided
```

```r
ks.test(X, "pnorm", 24.5, 2) # guess - did not detect a difference
```

```
##
##  Exact one-sample Kolmogorov-Smirnov test
##
## data:  X
## D = 0.095195, p-value = 0.7195
## alternative hypothesis: two-sided
```

```r
ks.test(X, "pnorm", 25.5, 2) # guess - did not detect a difference
```

```
##
##  Exact one-sample Kolmogorov-Smirnov test
##
## data:  X
## D = 0.21963, p-value = 0.01343
## alternative hypothesis: two-sided
```

```r
ks.test(X, "pnorm", 25, 5) # guess - detected a difference
```

```
##
##  Exact one-sample Kolmogorov-Smirnov test
##
## data:  X
## D = 0.25433, p-value = 0.002444
## alternative hypothesis: two-sided
```

```r
ks.test(X, "pnorm", 25, 2.5) # guess - did not detect a difference
```

```
##
##  Exact one-sample Kolmogorov-Smirnov test
##
## data:  X
## D = 0.12076, p-value = 0.4259
## alternative hypothesis: two-sided
```

Basically, if the mean was within 0.5 of the real mean, and same thing with SD, we had issues detecting differences. Overall though, this is pretty good news for identifying distributions. These are fairly small differences, and so we wouldn't be surprised to miss them.

Real Data Examples

Iris

```r
data(iris)
setosa <- filter(iris, Species == "setosa")
```

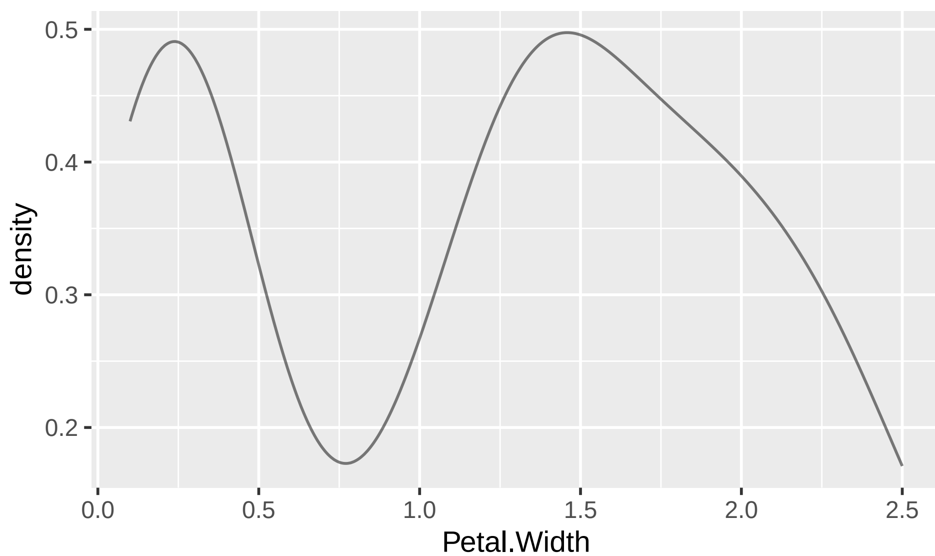
```r
virginica <- filter(iris, Species == "virginica")
versicolor <- filter(iris, Species == "versicolor")
```

We briefly return to the iris data set. Two variables weren't examined in the examples - Petal.Width and Sepal.Length.

Is the overall distribution of Petal.Width approximately normally distributed? If so, with what mean and SD? Use plots and a test to support your response.

SOLUTION:

```r
gf_dens(~ Petal.Width, data = iris) # expect it says this is NOT normally distributed
```




```r
favstats(~ Petal.Width, data = iris) # pull out mean and SD
```

```
##  min  Q1 median  Q3 max     mean        sd   n missing
##  0.1 0.3    1.3 1.8 2.5 1.199333 0.7622377 150       0
```

```r
mean(~ Petal.Width, data = iris) # could also pull out this way and save to input below
```

```
## [1] 1.199333
```

```r
sd(~ Petal.Width, data = iris)
```

```
## [1] 0.7622377
```

```r
ks.test(iris$Petal.Width,"pnorm", 1.19933, 0.762238)
```

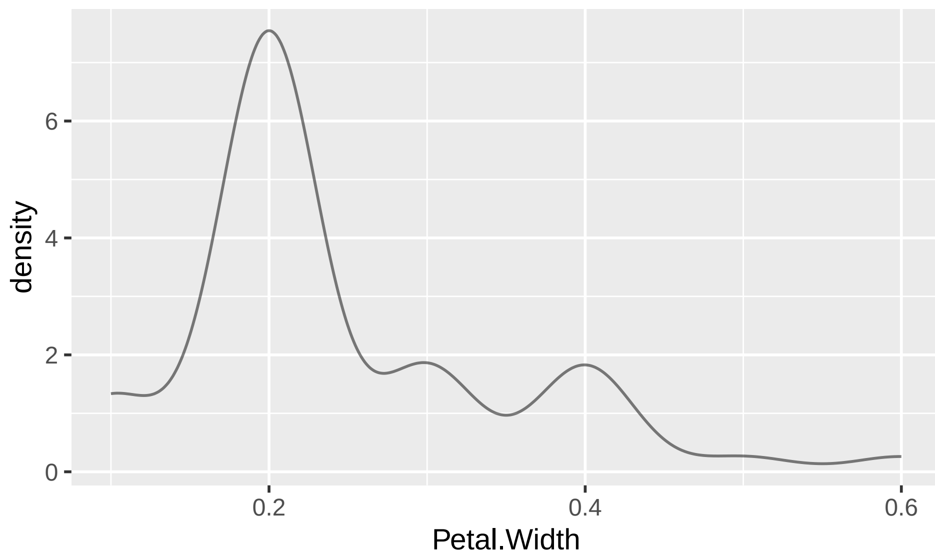
```
## Warning in ks.test.default(iris$Petal.Width, "pnorm", 1.19933, 0.762238): ties
## should not be present for the one-sample Kolmogorov-Smirnov test

##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  iris$Petal.Width
## D = 0.17283, p-value = 0.0002565
## alternative hypothesis: two-sided
```

The plots and small p-value from the KS-test suggest that the overall distribution of Petal.Width is not normally distributed (using the MLEs for mean and SD as the estimates of the distribution parameters).

Is the distribution of Petal.Width for just Setosa approximately normally distributed? If so, with what mean and SD? Use plots and a test to support your response.

SOLUTION:

```
gf_dens(~ Petal.Width, data = setosa)
```




```
# could be normal with some outliers, though second peak in upper tail
favstats(~ Petal.Width, data = setosa) # pull out mean and SD
```

```
##  min  Q1 median  Q3 max  mean        sd  n missing
##  0.1 0.2    0.2 0.3 0.6 0.246 0.1053856 50       0
```

```
mean(~ Petal.Width, data = setosa) # could pull out this way and save to input below
```

```
## [1] 0.246
```

```
sd(~ Petal.Width, data = setosa)
```

```
## [1] 0.1053856
```

```
ks.test(setosa$Petal.Width, "pnorm", 0.246, 0.105386)
```

```
## Warning in ks.test.default(setosa$Petal.Width, "pnorm", 0.246, 0.105386): ties
## should not be present for the one-sample Kolmogorov-Smirnov test
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  setosa$Petal.Width
## D = 0.34876, p-value = 1.044e-05
## alternative hypothesis: two-sided
```

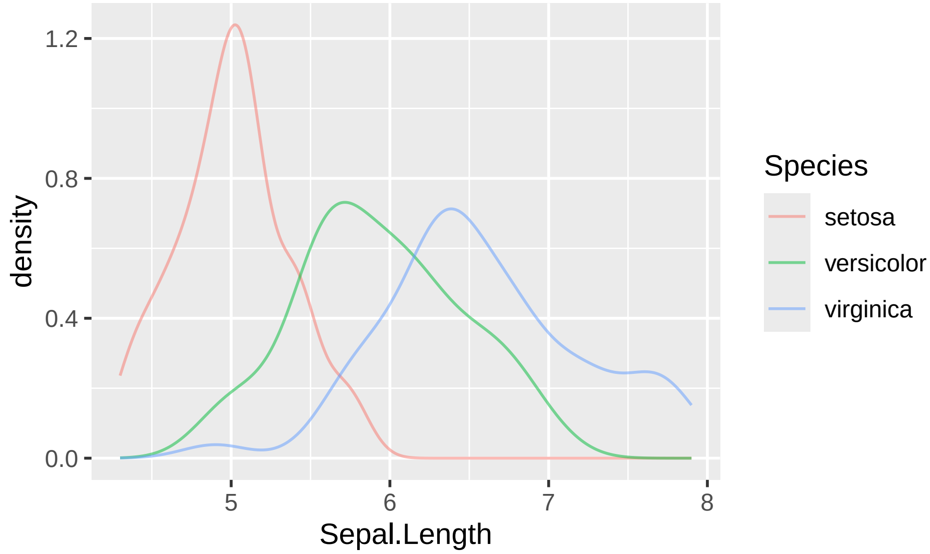
This rejects the null with a small p-value. Thus, it does not appear that the distribution of Petal.Width is normally distributed for Setosa.

Finally, we take a look at Sepal.Length and compare across the Species.

Is there evidence that any pair of Species has different distributions for their Sepal.Lengths? Use plots and a test to support your response.

SOLUTION:

```r
gf_dens(~ Sepal.Length, color = ~ Species, data = iris)
```



The setosa distribution looks different from the other two, shifted down a bit with less spread. The other two distributions are very similar in shape and spread, but virginica is shifted up above versicolor. We'd expect all three of these tests to reject the null - they are different distributions.

```r
ks.test(versicolor$Sepal.Length, virginica$Sepal.Length)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  versicolor$Sepal.Length and virginica$Sepal.Length
## D = 0.46, p-value = 1.868e-05
## alternative hypothesis: two-sided
```

```r
# look similar but virginica is shifted up a bit
ks.test(versicolor$Sepal.Length, setosa$Sepal.Length)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  versicolor$Sepal.Length and setosa$Sepal.Length
## D = 0.78, p-value = 1.034e-15
## alternative hypothesis: two-sided
```

```r
ks.test(setosa$Sepal.Length, virginica$Sepal.Length)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  setosa$Sepal.Length and virginica$Sepal.Length
## D = 0.92, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

The tests confirm our suspicions. We would conclude that all three distributions are different from one another.

Egyptian Skulls

The next set of examples use a data set on Egyptian skulls from 3 different time periods (period value of 1=4000 BC, 2=3300 BC, and 3=1850 BC). Note that period is currently numeric, but could be converted to a factor, as shown:

```
skulls <-
  read.table("https://awagaman.people.amherst.edu/nonparametricstats/egyptskulls.txt",
             h = T)
str(skulls)
```

```
## 'data.frame':    90 obs. of  5 variables:
##  $ breadth : int  131 125 131 119 136 138 139 125 131 134 ...
##  $ bbheight: int  138 131 132 132 143 137 130 136 134 134 ...
##  $ bvlength: int  89 92 99 96 100 89 108 93 102 99 ...
##  $ nasal   : int  49 48 50 44 54 56 48 48 51 51 ...
##  $ period  : int  1 1 1 1 1 1 1 1 1 1 ...
```

```
skulls <- mutate(skulls, periodf = factor(period)) # used a new name to not overwrite
str(skulls)
```

```
## 'data.frame':    90 obs. of  6 variables:
##  $ breadth : int  131 125 131 119 136 138 139 125 131 134 ...
##  $ bbheight: int  138 131 132 132 143 137 130 136 134 134 ...
##  $ bvlength: int  89 92 99 96 100 89 108 93 102 99 ...
##  $ nasal   : int  49 48 50 44 54 56 48 48 51 51 ...
##  $ period  : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ periodf : Factor w/ 3 levels "1","2","3": 1 1 1 1 1 1 1 1 1 1 ...
```

Measurements are all in mm for maximum breadth of skull (mm), basibregmatic height (bbheight), basialveolar length (bvlength), and nasal height (nasal). Briefly, we consider the individual variables for addressing the main question of interest.

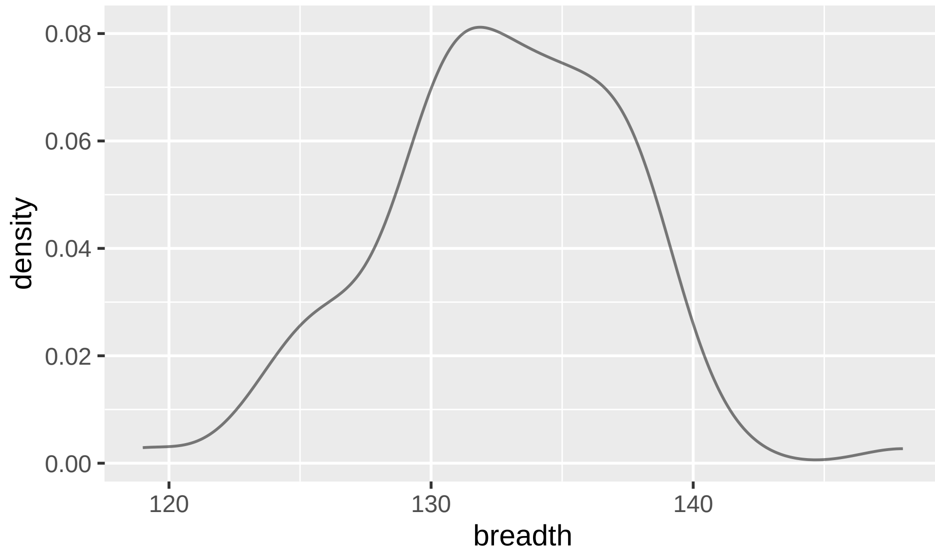
Is the distribution of breadth approximately normally distributed? If so, with what mean and SD? Use plots and a test to support your response.

SOLUTION:

```
favstats(~ breadth, data = skulls)
```

```
##  min  Q1 median  Q3 max     mean       sd  n missing
##  119 130    133 136 148 132.7333 4.663509 90       0
```

```
gf_dens(~ breadth, data = skulls)
```




```r
ks.test(skulls$breadth, "pnorm", 132.733, 4.66)
```

```
## Warning in ks.test.default(skulls$breadth, "pnorm", 132.733, 4.66): ties should
## not be present for the one-sample Kolmogorov-Smirnov test

##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  skulls$breadth
## D = 0.073628, p-value = 0.7137
## alternative hypothesis: two-sided
```

The `KS.test` gives us a large p-value, so we do not reject the null. We do not have significant evidence to suggest the distribution of breadth is different from a Normal(132.733, 4.66), where the mean and SD were estimated from the data.

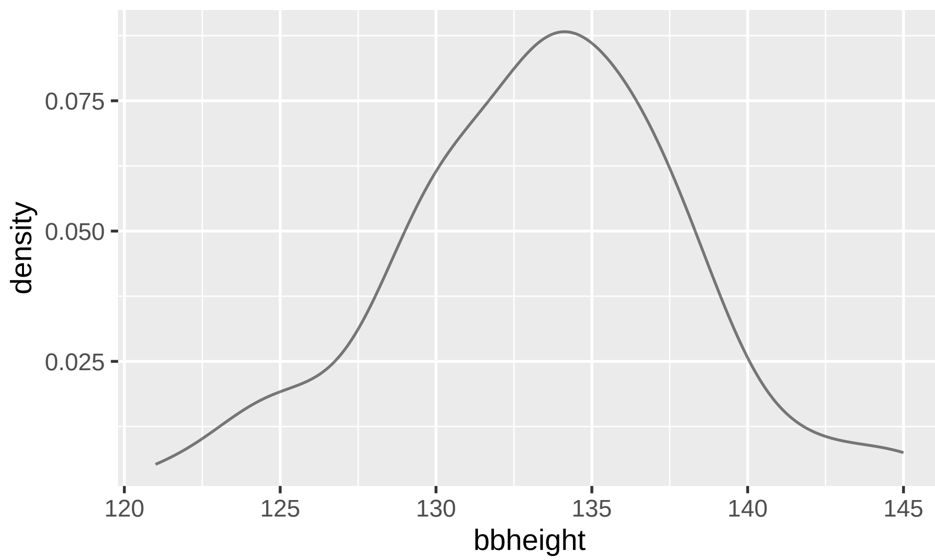
Is the distribution of bbheight approximately normally distributed? If so, with what mean and SD? Use plots and a test to support your response.

SOLUTION:

```r
favstats(~ bbheight, data = skulls)
```

```
##  min     Q1 median  Q3 max     mean       sd  n missing
##  121 130.25    134 136 145 133.3667 4.674699 90       0
```

```r
gf_dens(~ bbheight, data = skulls)
```




```
ks.test(skulls$bbheight, "pnorm", 133.367, 4.6747)
```

```
## Warning in ks.test.default(skulls$bbheight, "pnorm", 133.367, 4.6747): ties
## should not be present for the one-sample Kolmogorov-Smirnov test
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  skulls$bbheight
## D = 0.079823, p-value = 0.6149
## alternative hypothesis: two-sided
```

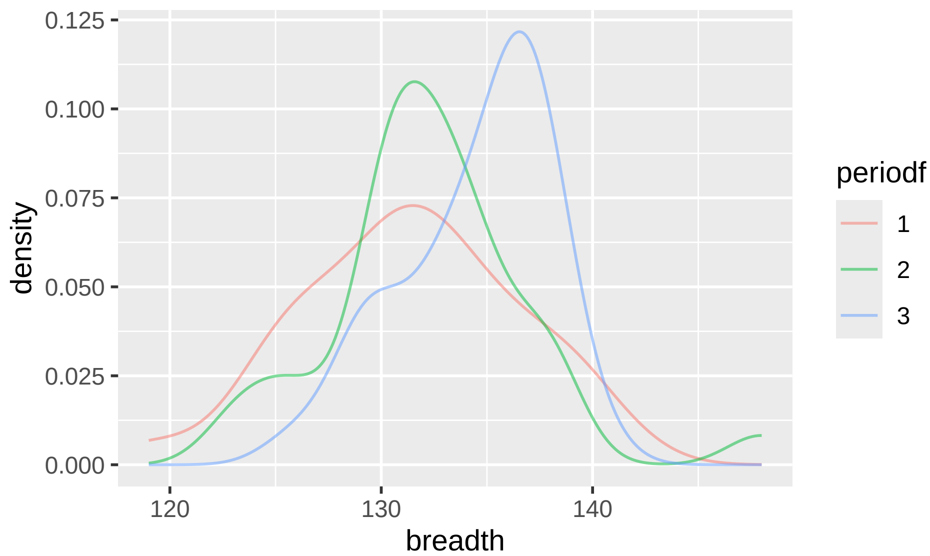
The `KS.test` again gives us a large p-value, so we do not reject the null. We do not have significant evidence to suggest the distribution of bbheight is different from a Normal(133.367, 4.6747), where the mean and SD were estimated from the data.

Changes in skull measurements over time may be indicative of breeding between native and immigrant populations, which is a question of interest to social scientists. Address the questions below - using both graphical methods (make pictures) and appropriate hypothesis tests.

Is there evidence that the breadth measurements from periods 1 and 3 (the periods most separated in time) come from different distributions? Use plots and a test to support your response.

SOLUTION:

```
gf_dens(~ breadth, color = ~ periodf, data = skulls) # p1 and p3 look different
```




```
p1 <- filter(skulls, period == 1)
p3 <- filter(skulls, period == 3)
ks.test(p1$breadth, p3$breadth)
```

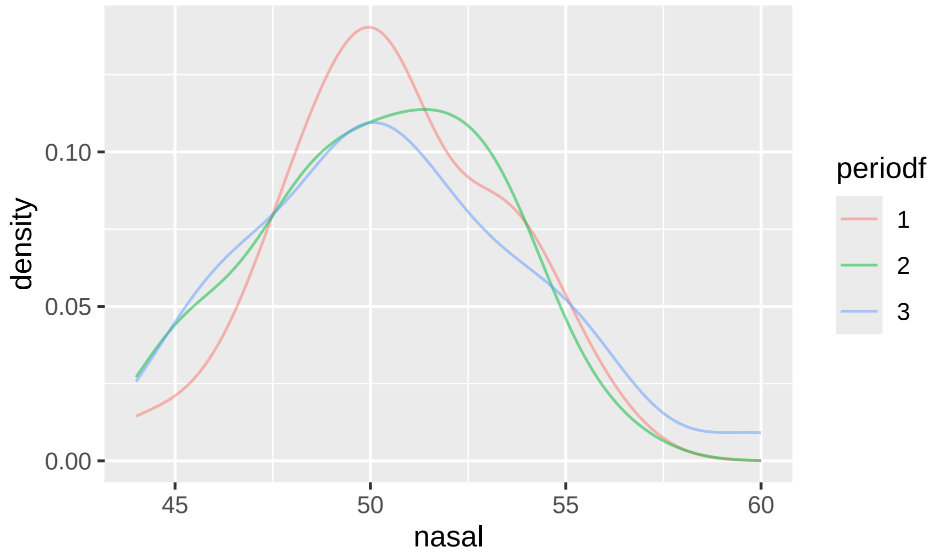
```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  p1$breadth and p3$breadth
## D = 0.36667, p-value = 0.02193
## alternative hypothesis: two-sided
```

From the density plot, we look at the red and blue curves. They appear to be different densities. So, we look at our KS test results. With a p-value of 0.02193 at alpha = 0.05, we reject the null. There is evidence to suggest the breadth distributions in period 1 and period 3 are different distributions.

Is there evidence that the nasal measurements from any pair of periods come from different distributions? Use plots and tests to support your response.

SOLUTION:

```
gf_dens(~ nasal, color = ~ periodf, data = skulls) # the distributions are all similar
```




```
p2 <- filter(skulls, period == 2)
ks.test(p1$nasal, p2$nasal)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  p1$nasal and p2$nasal
## D = 0.066667, p-value = 0.9995
## alternative hypothesis: two-sided
```

```
ks.test(p1$nasal, p3$nasal)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  p1$nasal and p3$nasal
## D = 0.13333, p-value = 0.7699
## alternative hypothesis: two-sided
```

```
ks.test(p2$nasal, p3$nasal)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  p2$nasal and p3$nasal
## D = 0.13333, p-value = 0.8612
## alternative hypothesis: two-sided
```

The density plot shows three VERY similar distributions. If any were going to be different, it would probably be period 1 versus the other two. We perform the three requested hypothesis tests, bearing in mind that we should use a low alpha for each test to avoid a multiple testing issue. The test results clearly indicate that we should not reject any of the null hypotheses. In conclusion, we do not have sufficient evidence to state that the nasal distributions are different across the three time periods.

Nonparametric Methods in Literature

As you learn new nonparametric methods, you are likely curious about how they are used in practice. Specifically, how are they used in application areas of interest to you? Let's explore how the Kolmogorov-

Smirnov test is used.

Use Google Scholar and do a search for “kolmogorov smirnov test” (or “KS test”) and an application area of interest to you. For example, search for ‘ “Kolmogorov Smirnov” archaeology’ to look for applications of the test in archaeology. (If the subject area has multiple words, you may want to put quotes around it as well.) Choose an article that pops up from your search that is interesting to you (more recent than 2010 would be best, if possible) and answer the questions below.

If you are having issues picking an application area or choosing an article, some potential articles are provided below. Applications do not need to be in journals in statistics!

Shaus, Arie, and Eli Turkel, “Writer Identification in Modern and Historical Documents via Binary Pixel Patterns, Kolmogorov–Smirnov Test and Fisher’s Method,” Electronic Imaging, 29, 2017: 203-211.

Monge, Marco, “Two-Sample Kolmogorov-Smirnov Tests as Causality Tests. A Narrative of Latin American inflation from 2020 to 2022,” Revista Chilena de Economía y Sociedad, 2023.

Bisgin, Halil, “Analyzing the attitudes of physical education and sport teachers towards technology,” The Anthropologist, 18.3, 2014: 761-764.

Provide a citation for your article. Google Scholar should be able to provide a complete citation for you to copy/paste.

SOLUTION:

Answers will vary.

Summarize what the article is about in a few sentences.

SOLUTION:

Answers will vary.

What was the KS test used for in the article? Describe as best you can. For example, what were the hypotheses being tested? Was it a one-sample or two-sample procedure?

SOLUTION:

Answers will vary.

Does the article state why the KS test was used? If so, what reason do they give?

SOLUTION:

Answers will vary.

Are any other statistical procedures mentioned in the paper, that you can tell? If so, what else is mentioned?

SOLUTION:

Answers will vary.

Data Sources

Iris data. Available in R with data(iris). Original sources referenced in R help:

Fisher, R. A. (1936) The use of multiple measurements in taxonomic problems. Annals of Eugenics, 7, Part II, 179–188. doi:10.1111/j.1469-1809.1936.tb02137.x.

Anderson, Edgar (1935). The irises of the Gaspe Peninsula, Bulletin of the American Iris Society, 59, 2–5.

The Egyptian skull dataset is a subset of a dataset from Thomson, A. and Randall-Maciver, R. (1905) Ancient Races of the Thebaid, Oxford: Oxford University Press. Also found in: Hand, D.J., et al. (1994) A Handbook of Small Data Sets, New York: Chapman & Hall, pp. 299-301. Manly, B.F.J. (1986) Multivariate Statistical Methods, New York: Chapman & Hall.

Standard Reference

These materials were created for use in an undergraduate nonparametrics course. Where provided, textbook references refer to Nonparametric Statistical Methods (3rd Edition) by Hollander, Wolfe, and Chicken. Notation is chosen to be consistent with that text. However, materials may be used independently of the textbook.

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